

David Nentwig

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Professional Summary

Computer Science graduate (UCL, 2:1) with strong Python expertise across data analysis, NLP, automation, and API integration. Experienced building end-to-end data pipelines, training large language models, and delivering data-driven solutions in professional and academic settings. Collaborative team player with agile experience and a track record of translating complex technical requirements into reliable, well-documented code.

Skills

Python	NumPy, pandas, scikit-learn, NLTK, Transformers (HuggingFace), PyTorch, PEFT, requests, web scraping, Plotly
Other Languages	C, C#, Java, Swift, JavaScript
Web	React, HTML5, CSS, component-based architecture
Tools	Git/GitHub, Docker, Poetry, pip, virtual environments, testing frameworks
Data & ML	SQL, LDA topic modelling, sentiment analysis, LLM fine-tuning (LoRA, 4-bit quantisation)
Languages	English (Native), German (Native), Welsh (Beginner), French (Beginner)

Experience

Resource Executive | Network ERP & IT (EMEA) | Cardiff, Wales 2025–Present

- Sourced and placed bilingual candidates across the DACH region, conducting end-to-end recruitment in English and German
- Managed candidate pipelines using RDB database system, maintaining accurate records and coordinating targeted outreach campaigns

Junior Data Scientist (Contractor) | DLT Science Foundation (Exponential Science) / Nodiens | London, UK 2024–2025

- Scraped and processed large volumes of social media data from Reddit and Telegram across 10 major cryptocurrencies using custom Python pipelines, producing structured datasets for downstream analysis
- Applied multiple statistical and NLP models to analyse sentiment and price dynamics, including Granger causality testing, lead-lag correlation analysis, and rolling window correlation across varying time horizons
- Built LDA topic modelling pipelines to identify and track trending discussion themes across cryptocurrency communities, with outputs published directly on the Nodiens platform as user-facing insights
- Co-authored a published market research report - Nodiens Mood Index and Price Dynamics (Issue 01-25, January 2025) - as a named Data Scientist alongside senior UCL researchers and PhD-level scientists

Technology Workshop Instructor | Young at Heart Forever (Charity), Rock UK Summit Centre | Trelewis, Wales 2018–2021

- Delivered technical training to elderly community members, developing strong communication skills for explaining complex concepts to non-technical audiences

Projects

Dissertation: Cryptocurrency Sentiment & Trust Analysis | Python, NLP, LDA, PyTorch, Transformers, LoRA

- Scraped and processed large volumes of social media posts from Mastodon across multiple cryptocurrencies using custom Python pipelines, producing structured JSON datasets for downstream analysis
- Performed sentiment analysis and LDA topic modelling, building trust-weighted scoring based on engagement metrics and follower ratios to filter low-quality data, and correlated sentiment trends with real cryptocurrency price data over time
- Fine-tuned a GPT-NeoX-20B LLM with LoRA adapters and 4-bit quantisation on the collected dataset, enabling natural language querying of cryptocurrency trend insights
- Built interactive visualisations using Plotly to communicate sentiment shifts and topic popularity over time

MotionInput | *UCL Systems Engineering Project with Intel, IBM & Microsoft* | *React, HTML5, CSS, JavaScript*

- Primary web developer for a UCL systems engineering project in collaboration with Intel, IBM and Microsoft
- Designed and developed the project website using React, maintaining comprehensive technical documentation and coordinating continuous updates throughout the project lifecycle
- Worked in an agile team environment with regular sprints, demos, and code reviews with industry partners

JavaTerminal | *UCL Learning Tool* | *Java, Docker, Git*

- Developed a BASH learning tool in a collaborative team, enabling users to experiment in a Docker-isolated environment without affecting personal files
- Implemented all core commands (echo, cut, grep, etc.) from scratch using test-driven development, writing full test suites for each feature to ensure correctness and reliability
- Contributed to code reviews and version control workflows, maintaining a well-documented and maintainable codebase

Personal Website | *React, JavaScript, CSS* | d-nentwig.github.io/my-website

- Built a responsive personal portfolio using React with component-based architecture for a maintainable, scalable codebase

Education

BSc Computer Science | **University College London (UCL)**

2021–2024

- Upper-Second Class Honours (2:1)

A Levels | **St David's Sixth Form College (Honours Student)**

2018–2021

- Mathematics (A*), Computer Science (A*), Psychology (A), German (A), Honours Program (Distinction)
- GCSEs: 1 A*, 8 As, 3 Bs